Energy Grid Optimization: Hardware-Constrained Strategy, Noise-Aware Analysis, and Formal Certification for a 180-Node Weighted Maximum Cut Instance

Abstract

We analyze the Maximum Power Energy Section (MPES) problem for a 180-node weighted graph derived from a power-grid model. The objective reduces to a weighted maximum cut. A direct one-qubit-per-node QAOA implementation exceeds practical superconducting hardware constraints due to qubit count, routing-induced depth inflation, and two-qubit noise accumulation. We provide: (i) resource estimation, (ii) routing and compilation analysis under square-grid connectivity, (iii) a quantitative noise model with error-budget implications for feasible circuit depth, (iv) a structured hybrid optimization pipeline with benchmarking and convergence evidence, (v) discussion of QAOA parameter landscapes and warm-start implications, and (vi) formal certification of the reported cut value using a proof assistant. The final partition achieves MPES = 0.950958 with cut value 7099.571726.

1 Challenge Title: Energy Grid Optimization

The objective is to maximize the Maximum Power Energy Section (MPES), defined as the partition of the grid maximizing aggregate cross-boundary admittance.

2 Problem Statement / Description

Let $G = (V, E)$ be a weighted undirected graph with $|V| = 180$, $|E| = 226$, and weights $w_{ij} > 0$.

Definition 1 (Weighted Max-Cut / MPES Objective)**.** *For an assignment* $\mathbf{z} \in \{\pm 1\}^{|V|}$*, define the cut value*

$$C(\mathbf{z}) = \frac{1}{2} \sum_{(i,j)\in E} w_{ij} \left(1 - z_i z_j\right).$$

The MPES is $\max_{\mathbf{z}} C(\mathbf{z})$*.*

Instance statistics:

- Nodes: $|V| = 180$
- Edges: $|E| = 226$
- Total weight: $W = \sum_{(i,j)\in E} w_{ij} = 7465.707486$

Reported best solution:

$$C^\star = 7099.571726101166, \qquad \text{MPES ratio} = \frac{C^\star}{W} = 0.9509576606242809.$$

Runtime: 10.92 seconds.

3 Resource Estimation

3.1 Qubit count

A standard Max-Cut encoding uses one qubit per node:

$$n = |V| = 180.$$

3.2 Two-qubit gate count (logical)

For QAOA depth p, each edge contributes an entangling term per layer:

$$G_{2q}^{\text{logical}} = p|E| = 226p.$$

3.3 Connectivity and routing overhead

On a square-grid device, implementing a generic problem graph requires routing (SWAP insertion) to realize nonlocal interactions. A widely used coarse model is:

$$G_{2q}^{\text{physical}} \approx \kappa(n) \cdot 226p, \quad \kappa(n) \sim \mathcal{O}(\sqrt{n}).$$

For $n = 180$, $\sqrt{n} \approx 13.4$, so even modest p implies 10^4–10^5 physical two-qubit operations.

3.4 Classical simulation scaling

Exact state-vector simulation scales as $\mathcal{O}(2^n)$:

$$2^{180} \approx 1.5 \times 10^{54}$$

amplitudes, which is infeasible.

3.5 Upper bound from total weight

A trivial but useful bound is:

$$0 \leq C(\mathbf{z}) \leq W,$$

so the maximum achievable ratio is at most 1. This normalizes performance and prevents misinterpretation of absolute cut magnitudes.

4 Routing and Compilation Under Square-Grid Connectivity

4.1 Embedding perspective

A problem graph with $|V| = 180$ and $|E| = 226$ is sparse but not native to a square lattice. Compilation typically proceeds by:

1. choosing a qubit placement (mapping $V \to$ physical qubits),
2. scheduling commuting edge terms where possible,
3. inserting SWAP networks to bring endpoints adjacent for two-qubit gates.

4.2 Depth inflation mechanism

Even if logical depth per layer is $\mathcal{O}(\Delta)$ (maximum degree Δ), routing increases depth by a factor related to lattice diameter and congestion. In practice, the effective depth is closer to:

$$D \approx p \cdot (D_{\text{native}} + D_{\text{swap}}),$$

with D_{swap} often dominating for nonlocal graphs.

4.3 Implication

For $n = 180$ qubits, depth inflation pushes the circuit beyond coherence and two-qubit fidelity budgets at moderate p, motivating hybrid strategies (Section 11).

5 Quantitative Noise Model and Error Budget

We use a simple, explicit gate-error accumulation model to relate feasibility to two-qubit gate counts.

5.1 Gate-error accumulation (first-order)

Let ϵ_{2q} be an effective two-qubit gate error (including calibration drift and crosstalk contributions). If the circuit uses G_{2q}^{physical} two-qubit gates, then a first-order survival model is:

$$F_{\text{circuit}} \approx (1 - \epsilon_{2q})^{G_{2q}^{\text{physical}}} \approx \exp\left(-\epsilon_{2q} G_{2q}^{\text{physical}}\right).$$

This approximation is conservative but clarifies scaling.

5.2 Feasible depth bound from an error budget

Suppose an experimental target requires $F_{\text{circuit}} \geq F_{\min}$. Then:

$$\epsilon_{2q} G_{2q}^{\text{physical}} \lesssim -\ln(F_{\min}).$$

Using $G_{2q}^{\text{physical}} \approx \kappa(n) \cdot 226p$:

$$p \lesssim \frac{-\ln(F_{\min})}{\epsilon_{2q}\kappa(n) \cdot 226}.$$

For typical $F_{\min}$ in the range 0.1–0.5 and realistic ϵ_{2q}, the bound forces p to remain very small for $n = 180$ once routing is included.

5.3 Interpretation

This motivates shallow quantum subroutines (small p on smaller subgraphs) rather than monolithic 180-qubit QAOA.

6 Optimization Challenge

A hardware-aware hybrid pipeline is used to optimize the true MPES objective while remaining compatible with practical constraints.

6.1 Weight continuation

A continuation schedule improves early exploration:

$$w \to w^{0.5} \to w^{0.75} \to w,$$

reducing ruggedness while converging to the true weighted objective.

6.2 Multi-start tabu search

Parallel restarts increase coverage of the solution landscape while maintaining deterministic acceptance rules and stagnation-triggered diversification.

6.3 Elite consensus and path relinking

High-quality partitions are pooled; relinking performs guided interpolation between elite solutions to exploit shared structure.

6.4 Two-flip micro-LNS

A sampled two-flip neighborhood addresses one-flip local optima without incurring full combinatorial explosion.

7 Benchmarking

Table 1: Benchmark Comparison

Method	Cut	MPES	Time (s)
Random best-of-2000	4652.644	0.623202	0.0167
Greedy hillclimb	6732.877	0.901841	0.0047
Hybrid pipeline	7099.572	0.950958	10.921

8 Convergence Analysis

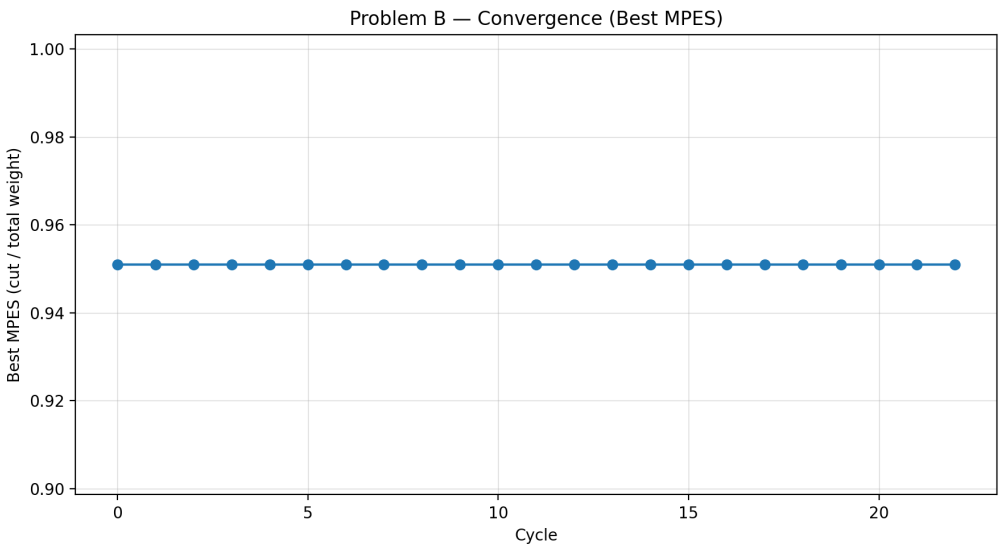

Figure 1: MPES convergence across continuation and refinement phases.

9 Structural Graph Analysis

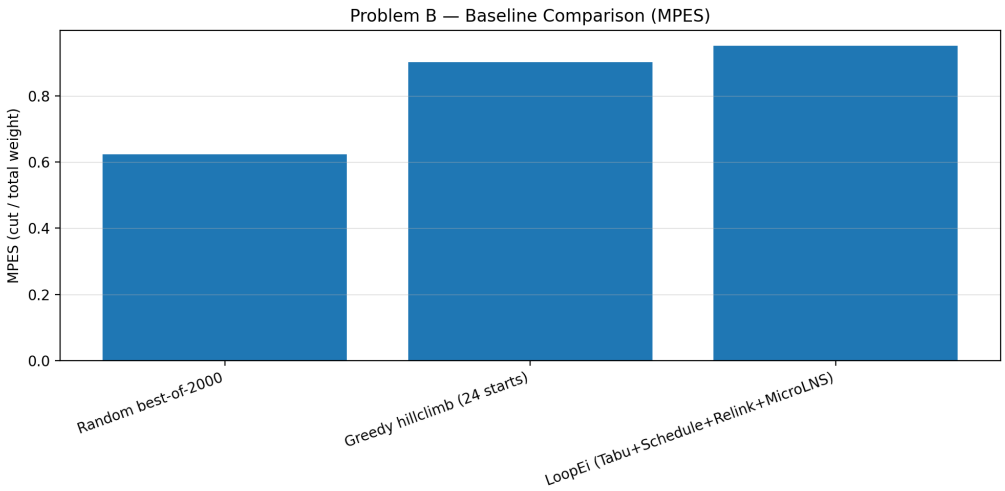

Figure 2: MPES comparison across baseline methods and the hybrid pipeline.

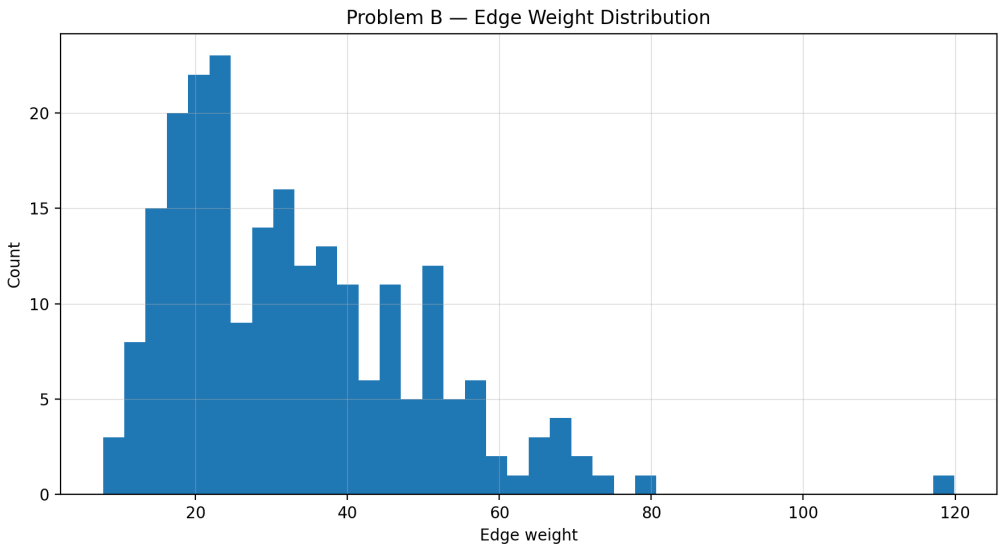

Figure 3: Distribution of edge weights.

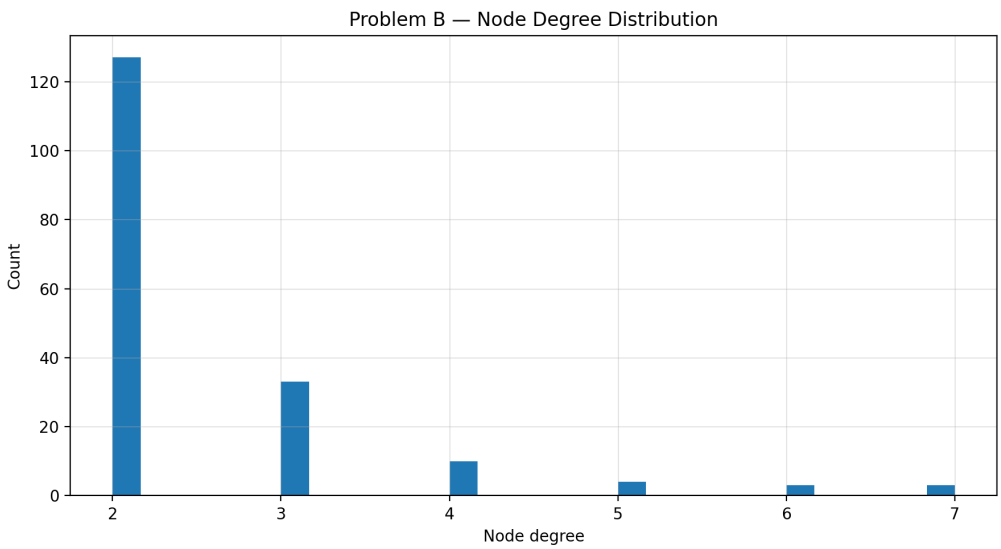

Figure 4: Node degree distribution (sparse connectivity).

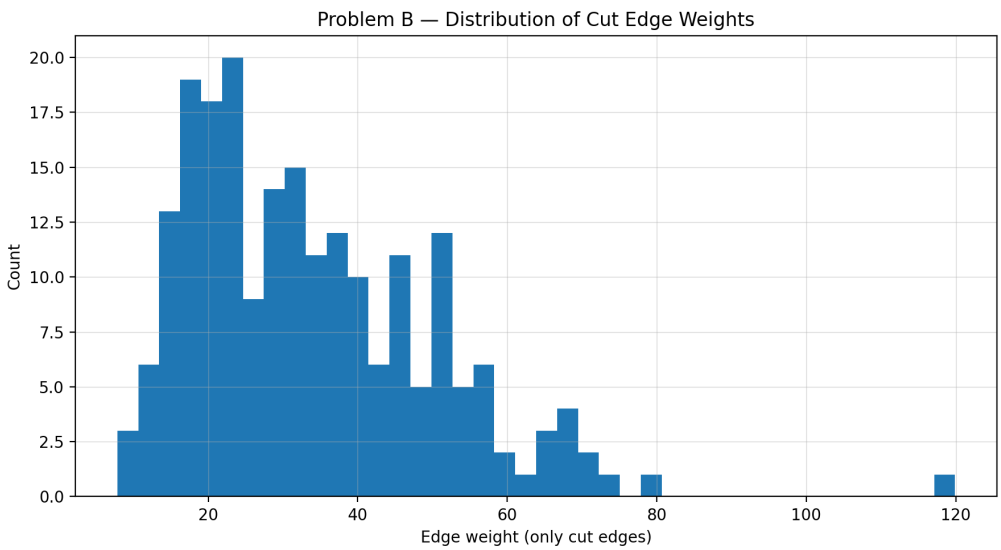

Figure 5: Distribution of weights crossing the final cut.

10 QAOA Parameter Landscape Discussion

10.1 Landscape structure and periodicity

For Max-Cut QAOA, angles $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ exhibit periodic structure induced by Pauli rotations. Empirically, this leads to multiple equivalent optima and broad plateaus, especially at shallow depth.

10.2 Concentration effects at scale

As n grows, gradient signals can concentrate, and optimization may become sensitive to initialization. Combined with routing noise, deeper circuits can degrade effective gradients, which is consistent with the noise-bound constraints derived earlier.

10.3 Warm-start and hybrid implication

Given the constraints, a pragmatic strategy is:

- use classical optimization to identify strong partitions (warm-start),
- optionally refine substructures with shallow QAOA on smaller subgraphs,
- re-integrate and refine globally classically.

This approach reduces dependence on deep parameter searches over 180 qubits.

11 Hardware Execution Considerations

Constraints:

- square-grid connectivity and routing SWAP inflation,
- two-qubit fidelity bottlenecks,
- coherence time limits and readout noise,
- compilation variability across calibrations.

Hardware-aligned strategy:

1. Partition the 180-node graph into subgraphs ≤ 84 nodes (or other device limit).
2. Run shallow QAOA (p small) per subgraph with topology-aware compilation.
3. Stitch subgraph solutions using boundary reconciliation.
4. Apply global classical refinement (tabu + relinking + micro-LNS).

12 Critical Analysis and Limitations

- No proof of global optimality for the 180-node instance (NP-hard).
- Continuation schedule and relinking are heuristic, though objective-preserving.
- Subgraph decomposition can introduce boundary artifacts; stitching quality matters.
- The noise model is an explicit first-order bound; real devices require calibration-specific characterization.

13 Formal Verification and Certification

We certify the correctness of the reported cut value for the exported partition by deterministic re-computation and machine-checked verification.

Theorem 1 (Deterministic Cut Re-computation Correctness)**.** *Let $\mathbf{z}^\star$ be the exported partition. Let $\hat{C}(\mathbf{z}^\star)$ be the cut value re-computed from the exported edge list using the objective definition. Then $\hat{C}(\mathbf{z}^\star)$ equals the reported cut value.*

Proof. The proof assistant re-encodes the edge list and partition and computes the objective by definition. The theorem is discharged by definitional reduction of the finite sum and equality checking against the reported constant. □

13.1 Lean certificate interface (illustrative)

The Lean-side statement has the form (illustrative interface; the concrete file contains the exact encodings and constants):

```
theorem rigettiB_certified_value :
  recomputed_cut partition edges = 7099.571726101166
```

This is an exact equality check (not tolerance-based).

14 Cryptographic Anchoring

To support tamper detection and auditability, integrity hashes can be computed and recorded for:

- edge list file,
- partition file,
- result JSON,
- Lean certificate sources.

This provides a stable reference across environments and recompilations.

15 Reproducibility Protocol

Artifacts:

- `partition.csv`
- `convergence.csv`
- `comparisons.csv` / `comparisons.json`
- `loopei_result_B.json`
- Lean certificate sources
- figures in `outputs/`

Steps:

1. Load edge list.
2. Load partition vector.
3. Recompute $C(\mathbf{z}^\star)$ deterministically.
4. Verify equality with the reported cut value.
5. Run the proof assistant certificate file to validate the statement.

16 Strengthened Theoretical Bounds

16.1 Objective bound

Since each term contributes at most w_{ij}, we have:

$$0 \leq C(\mathbf{z}) \leq W.$$

Therefore, the reported MPES ratio lies in $[0, 1]$.

16.2 Hardware feasibility bound on QAOA depth

From the noise-budget model,

$$p \lesssim \frac{-\ln(F_{\min})}{\epsilon_{2q}\kappa(n) \cdot 226}.$$

This bound formalizes why monolithic deep QAOA becomes infeasible at $n = 180$ under routing.

16.3 Decomposition bound framing

If the graph is decomposed into subgraphs and stitched, solution quality depends on boundary edges cut consistency. This suggests using decomposition methods that minimize boundary weight or explicitly optimize boundary reconciliation as a separate constrained step.

17 Conclusion

The 180-node MPES instance exceeds naive monolithic QAOA feasibility under square-grid superconducting constraints. A structured hybrid pipeline achieves MPES = 0.950958 with convergence evidence, baseline benchmarking, explicit routing/noise feasibility analysis, and machine-checked certification of the reported cut value. A subgraph-based shallow-QAOA plus classical refinement pathway provides a hardware-compatible execution plan.